

- 7.19. Explain why the increased securitization of commercial real estate, on both the debt and equity sides, could make real estate markets less susceptible to boom and bust cycles. What happened in 1998 that helps support this line of reasoning?
- 7.20. [Aging of the Baby Boom and the Asset Demand for Real Estate] Given what you now know about investor objectives and constraints, the historical performance of the major investment asset classes, as well as the more mainstream acceptance of real estate investment in recent years, think about the impact of the aging baby boom generation on investor demand for real estate going forward. Do you think the movement of boomers into retirement age implies that the real estate asset class will continue to attract significant capital relative to other asset classes? Support your answer with sound economic reasoning, based on what you learned in the chapter.

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8.3 Chapter Summary

(Continued)

Present Value Mathematics for Real Estate

Learning Objectives

After reading this chapter, you should understand:

- The basic formulas and procedures for converting typical real estate future cash flow and risk patterns to present value, given the appropriate discount rates.
- The origin or derivation of these formulas, in such a way that you can apply them with some flexibility to new situations.
- How to invert the present value formulas to compute the implied return or the time until a future value is received.
- How a typical business or financial calculator works to make these calculations quick and easy.

Chapter Outline On the Accompanying CD:

Business
Calculator
Keystrokes &
Excel®
Examples

If someone offered you a choice between \$10,000 today or \$10,000 one year from now, which would you choose? It is a fundamental fact of financial economics that dollars at one time are not equivalent to dollars at another time. This is not just because of inflation. Due to the real productivity of capital, and due to risk, future dollars are worth less than present dollars, even if there were no inflation. In real estate investment analysis, we constantly need to compare dollars at different points in time. Present value mathematics consists of the formulas, procedures, and techniques for making such comparisons and evaluations. This chapter will present some of the most basic and commonly used of these tools.

Please keep in mind that the focus of this chapter is on the mechanics, the formal mathematical formulas. In order to know how to use these formulas properly in real estate investment analysis, you need to know the right numbers to plug into the formulas. This requires combining economics with mathematics, a combination we will explore in later chapters.

For the purposes of this chapter, present value formulas can be grouped into two classes: single-sums and multiperiod cash flows. The single-sum formulas are the most basic building blocks, as the multiperiod formulas are all built up as sums of the single-sum formulas. So we will begin this chapter with single-sum formulas, and then advance to multiperiod formulas in the second half.¹

8.1 SINGLE-SUM FORMULAS

¹ Single-sum formulas refer to situations in which only two points in time are considered. Typically, one of these two points is the present. A single cash flow or monetary value is posited at one of the two points in time, and the task is to determine the value it equates to at the other point in time.

8.1.1 Single-Period Discounting and Growing

The simplest sort of present value problem is single-period discounting or growing. Let's go back to the choice between \$10,000 today versus one year from today. We no doubt agree that we would all choose the \$10,000 today. But suppose the choice is \$10,000 today versus \$11,000 in one year, or \$12,000, and so on. Clearly, there is some value greater than \$10,000 that we would prefer to take in one year. How high that "indifference point" depends on several things. For example, we would want to consider how much interest we could earn on a one-year deposit in a bank. If we could earn 5% simple interest on a one-year CD, then \$10,500 in one year would be equivalent to \$10,000 today, because $10,000 \times 1.05 = 10,500$. However, we might require a promise of more than \$10,500 in a year in exchange for our \$10,000 today if we view the future promise as being more risky than a bank CD. In effect, we might want more than a 5% expected return to get us to forgo the present cash. Suppose we wanted a 10% expected return "risk premium" on top of the 5% we could earn at the bank, for a total expected return of 15% in order to entice us to forgo our \$10,000 today in exchange for a cash sum promised next year. Then \$11,500 is our indifference point, because

$$\$10,000 + (15\% \times \$10,000) = \$11,500$$

or equivalently

$$\$10,000 \times (1 + 0.15) = \$11,500$$

¹If you are already familiar with these formulas and procedures, you can safely skip this chapter. The only thing we do here that is a little different from the typical coverage is that we present the underlying geometric series formula from which the regular multiperiod (annuity and perpetuity) present value formulas are derived. This will give the reader some additional depth and flexibility and enable us to show how to develop formulas for the present values of buildings with long-term leases, a common problem in commercial real estate.

Now suppose that we know our required expected return percentage (say, 15%, as in the previous example), and we know the amount that is being promised one year from now. So, the question is, how much are we willing to forgo today in exchange for the promised sum in one year? We can find this present value by simply inverting the previous formula and solving for the present value rather than for the future value. For example, if \$11,500 is being offered in one year, then we could determine that we are willing to pay \$10,000 today by dividing the expected future amount by one plus the expected return, as follows:

$$\$10,000 = \frac{\$11,500}{1 + 0.15}$$

Assuming that 15% is our required expected return (opportunity cost of capital), any offer of more than \$11,500 next year would be preferred over holding on to our \$10,000 today, while any offer of less than \$11,500 would be rejected in favor of keeping our \$10,000 today.

In the previous example, the \$10,000 amount is the *present certainty equivalent*, often more simply referred to as the **present value**, of the *future expected sum* (or **future value**) of \$11,500 in one year, based on a simple annual expected return of 15%. In general, if we let *PV* represent the present value, *FV* represent the future value, and *r* represent the simple expected return, then the formula for discounting future values to present value is:

$$PV = \frac{FV}{1 + r} \quad (1a)$$

and the corresponding formula for growing a present value to its future equivalent is:²

$$FV = (1 + r)PV \quad (1b)$$

Using formula (1a), you could answer the question of how much cash in the present would equate to \$10,000 promised money in one year, under the 15% required return assumption of the previous numerical example. The answer would be $PV = \$8,695.65 = 10,000/1.15$. When solving for present value given the future value, the problem is one of **discounting** rather than **growing**, and the required expected return percentage in the denominator acts as the **discount rate**. Discounting is merely the inverse of growing.

8.1.2 Single-Sums over Multiple Periods of Time

The problem in the previous section involved only a single period of time, with the return or discount rate expressed as a simple rate per the amount of time in that period. For example, if everything in the earlier numerical example were the same except that the waiting period between the present cash and the future payment were two years instead of one year, then the formulas would still apply only we would have to interpret the 15% return as a simple biennial rather than annual return, that is, a return over two years instead of one year.

Now suppose there is more than one period of time between the time of the *PV* amount and the time of the *FV* amount, yet we want to express the return per unit of time in each period. For example, suppose the future payment is promised in two years and we want to obtain 15% simple annual return *each* year. If you put, say, \$100 in a one-year CD at 15% simple annual interest, you would have \$115 ($1.15 \times \100) after one year. If you took that \$115 and put it in another 15% CD again the

²This is the first of many formulas you will encounter in this chapter. Most people don't like memorizing formulas, so business calculators are sold that effectively "translate" these formulas into keystrokes that are entered in a pretty intuitive manner. We have included the keystrokes for business calculators for all of the problems presented in this chapter, in the PowerPoint® Lecture Notes file for this chapter available on the CD accompanying this book. The CD also has an Excel® file for this chapter with the formulas for all of the odd-numbered study questions at the end of this chapter, which cover all of the types of problems presented in the chapter.

second year, your \$115 would grow to \$132.25 ($1.15 \times \$115$) after the second year. In other words, \$100 growing at 15% simple annual interest compounded annually for two years is

$$(1.15)(1.15)100 = (1.15)^2 100 = 132.25$$

Note that this is more than the \$130 you would have if you simply took the original \$15 interest and multiplied by two. This is because you earn interest on interest (or return on return) whenever the interest or return is compounded (that is, computed and credited to the principal in the account). This process is known as **compounding** or **compound growth**.

It is easy to see by extension of this reasoning that if we were dealing with three years instead of two, we would have

$$(1.15)(1.15)(1.15)100 = (1.15)^3 100 = 152.09$$

So, by inductive reasoning, we can see that the general formula for equating present and future value across multiple periods with compounding once per period is given by

$$PV = \frac{FV}{(1+r)^N} \quad (2a)$$

for discounting a future sum to present value and:

$$FV = (1+r)^N PV \quad (2b)$$

for compounding a present sum forward to its future equivalent, where r is the simple interest or discount rate (or expected return) *per period*, and N is the number of periods between the present and the future time when the future sum is to be received.

The previous formulas are the building blocks for classical present value and mortgage mathematics. In modern practice, these formulas are normally applied using an electronic calculator or computer spreadsheet. For example, in a typical computer spreadsheet (such as Lotus or Excel), the right-hand side of (2a) would be keyed in as $FV/(1+r)^N$, where FV , PV , r , and N represent the cell addresses of the relevant future sum, present value, simple interest rate per period, and number of periods, respectively.

In case you need to solve for the compound growth rate or the length of time until the future value will be received, the previous formulas can be algebraically inverted to solve for r or N :

$$r = \left(\frac{FV}{PV} \right)^{1/N} - 1 \quad (2c)$$

$$N = \frac{\ln(FV) - \ln(PV)}{\ln(1+r)} \quad (2d)$$

where $\ln(\cdot)$ refers to the natural log. Of course, modern calculators and computer spreadsheets allow these types of problems to be solved without your actually having to remember formulas such as these!

8.1.3 Simple versus Compound Interest

From the discussion in the previous section, you can see that there is more than one way to define how the interest rate (or equivalently, the rate of return, or the discount rate) is measured per period when there is more than one period between the PV cash flow and the FV cash flow. For example, go back to the example in the previous section in which we started out with \$100 in the present and grew this to \$132.25 after two years. We saw how this growth could be expressed as a per annum

(or annual) rate of 15%, using annual compounding, because $(1 + 0.15) \times (1 + 0.15) = 1.3225$. But it would also be possible to take the total return increment over the two years, the 32.25%, and simply divide this by two and express the return as 16.125% per annum. This latter definition is called simple interest, because it does not include the effect of compounding.

The important point to remember is that how we define the interest or return per period does not affect the actual dollars that we have at the two points in time. The definition is merely a matter of convention. Whether we call it 15% per year compounded annually, or 16.125% per year simple interest, it is still \$132.25 starting out from \$100 two years earlier. You do have to be careful to understand the definition being used in a given context, and you must be consistent (e.g., don't compound simple interest in formulas to compute future sums).

8.1.4 Effective versus Nominal Rates

In the example in the previous section the compound interest rate was computed assuming the interest was compounded once per year (annual compounding). But, in fact, interest may be compounded at any **frequency**. Often, interest is compounded more frequently than once per year. For example, suppose 7.5% semiannual interest were compounded over a year (i.e., twice). After one year, you would have $(1.075)(1.075) = 1.155625$ times whatever amount you started with. As you can see, 7.5% compounded semiannually would be equivalent to a little over 15.56% simple annual interest or interest compounded annually. Nevertheless, in terms of the conventions of measurement and quoting of interest rates, the 7.5% compounded semiannually would usually be referred to as 15% per annum interest, compounded semiannually. The term **effective annual rate** (or **EAR**) would be used to refer to the 15.5625% actual yield.

In real estate applications, cash flows often occur at monthly intervals. For example, rents and mortgage payments are usually scheduled at monthly intervals. As a result, monthly interest rates are often actually applied in calculations, but rates are quoted in what is called **nominal per annum terms**. Under the convention of quoting rates in nominal per annum terms, the simple per-period rate (which is the rate that is actually applied in calculations) is multiplied by the number of periods in a year for the purpose of quoting the rate in per annum terms. For example, a simple interest rate of 1% per month would typically be referred to as a **12% per annum rate** (i.e., the nominal per annum rate), even though the effective rate or EAR is $(1.01)^{12} - 1 = 12.6825\%$. The EAR is defined as the actual annual yield corresponding to the nominal rate defined previously, given the simple per-period rate compounded at the end of each period. The relationship between effective and nominal rates is shown in the following formulas:

$$EAR = (1 + i/m)^m - 1 \quad (3a)$$

$$i = m[(1 + EAR)^{1/m} - 1] \quad (3b)$$

where i is the nominal annual rate, m is the number of compounding periods per year, and $i/m = r$ is the actual simple interest rate being applied per compounding period. In the previous example, we had $i = 12.00\%$, $EAR = 12.6825\%$, $r = 1.00\%$, and $m = 12$. The nominal annual rate i is sometimes referred to as the **equivalent nominal annual rate (ENAR)**.

As a result of this naming convention, when applying the present value and future value formula (2) described previously, one usually works in nominal terms by expanding the simple per-period interest rate compounded once per-period (r) and the number of basic periods of time (N) as follows, consistent with the earlier definitions:

$$r = i/m$$

$$N = m \times T$$

where i is the nominal annual rate of interest (e.g., 12% in the previous example), m is the number of compounding periods per year (e.g., 12 in the example of monthly compounding), and T is the length of time between when the PV and FV cash flows occur *measured in years*. Recast in nominal per-annum terms (allowing for a compounding frequency different from yearly), the present value formulas for single-sums become as follows:

$$PV = \frac{FV}{\left(1 + \frac{i}{m}\right)^{mT}} \quad (4a)$$

$$FV = PV \left(1 + \frac{i}{m}\right)^{mT} \quad (4b)$$

$$i = m \left[\left(\frac{FV}{PV} \right)^{1/(mT)} - 1 \right] \quad (4c)$$

$$T = \frac{\ln(FV) - \ln(PV)}{m \ln(1 + i/m)} \quad (4d)$$

Most calculators designed especially for business applications allow the user to set the value m so the calculator knows the number of compounding periods per year, and the user can then enter the interest rate in its nominal annual form (that is, you enter i and T rather than r and N ; e.g., in the previous example you would enter 12 for the interest rate rather than 1 or .01, and enter the number of years between the PV and FV cash flows, rather than the number of months). The calculator then automatically divides i by m to determine the rate per period to apply in the formulas. On the other hand, microcomputer spreadsheet financial formulas typically require as direct inputs the rate per period (r) and the number of periods (N). In order to work in nominal annual terms, the user must therefore have the spreadsheet calculate r and N from i , m , and T in other cells of the worksheet, and have these other cells serve as the inputs to the present value formulas.

8.1.5 Bond Equivalent and Mortgage Equivalent Rates

As noted, rent is usually collected monthly and mortgage payments are usually made monthly, so nominal rates (i , or the ENAR, using the earlier labels) based on the previous formulas with $m = 12$ is the most common way interest rates or yields are quoted in real estate. Nominal rates defined this way are sometimes referred to as **mortgage equivalent rates**.

Meanwhile, over on Wall Street, traditional corporate bonds and long-term government bonds are designed to pay interest semiannually, that is, twice per year.³ In fact, classical bonds had coupons corresponding to each of the promised semiannual interest payments, which the bondholder would clip off and turn in at a bank to receive the payment in cash. Because of this, in the bond market, rates are typically quoted with $m = 2$ in formula (3). Nominal rates based on $m = 2$ are thus referred to as **bond equivalent** or **coupon equivalent rates**.

Thus, real estate rates and bond rates are quoted in different ways. This was fine as long as Wall Street and Main Street did not interact very directly. But nowadays many mortgages are combined in pools and sold in the bond market, and bond market yields largely determine mortgage market yields. People dealing with mortgages need to be able to convert readily between mortgage equivalent and bond equivalent yields. This is done by *equating the effective annual rate (EAR)*. For

³The terms *semiannual* and *biannual* are equivalent, both referring to twice per year. The term *biennial* is different, referring to once every two years.

example, we noted that a bond equivalent rate of 15% corresponds to an EAR of 15.5625%, found by plugging $i = 0.15$ and $m = 2$ into formula (3a). Now plug this 15.5625% EAR into formula (3b) with $m = 12$ to find that the mortgage equivalent rate of 14.551655% corresponds to the bond equivalent rate of 15%. Mortgage equivalent rates are lower than their corresponding bond equivalent rates because the increased frequency of compounding results in faster growth. If the mortgage equivalent rate were 15%, this would correspond to a bond equivalent rate of 15.4766% and an EAR of 16.07545%. [Plug $i = 0.15$ and $m = 12$ into formula (3a) to obtain the EAR, which you then plug into formula (3b) with $m = 2$.] As always, this type of operation is easily done using financial calculators by changing the value in the "periods per year" register, without the need to memorize the formulas. You simply substitute the appropriate value for m in the "periods per year" register. (See Questions 8.11 through 8.20 at the end of this chapter.)

*8.1.6 Continuously Compounded Interest Rates

The extreme limit of frequent compounding of interest is **continuous compounding**. Many bank accounts, CDs, and other financial instruments employ continuous compounding, and many interest rates are quoted in continuously compounded terms. Continuous compounding is also mathematically convenient for many types of analyses, especially those involving advanced mathematics, and so is often used in sophisticated studies.⁴ In continuous compounding, the interest is literally compounded every instant.

Computing continuously compounded interest rates and continuously compounded growth or discounting is easy using the natural logarithm and the base of natural logs (the latter is known by the mathematical symbol e , which stands for a value approximately equal to 2.7183, and is available as a special key on most business calculators and in all spreadsheet software). The continuously compounded equivalents corresponding to the various forms of formulas (2) or (4) are shown as formula (5), where T is the amount of time in years (including fractional parts of a year) between the PV and FV cash flows, and k is the *per-annum* continuously compounded rate (of growth or discount, i.e., the continuously compounded interest rate).

$$PV = FV / e^{Tk} \quad (5a)$$

$$FV = PV e^{Tk} \quad (5b)$$

$$k = \frac{\ln(FV) - \ln(PV)}{T} \quad (5c)$$

$$T = \frac{\ln(FV) - \ln(PV)}{k} \quad (5d)$$

8.2 MULTIPERIOD PROBLEMS

The formulas described in Section 8.1 are the basic building blocks. They deal with only one future cash flow. Of course, real estate is a long-lived asset that typically produces cash flows for many periods, or indeed infinitely into the future. The present value of any stream of discrete cash flows

⁴Continuous compounding enables one to convert complex multiplicative expressions into simple additive expressions using natural logs. Additive expressions are much easier to work with mathematically. For example, if r_1 is the simple rate of return in period t , then at the end of two periods, $FV = (1 + r_1)(1 + r_2)PV$, but if r_1 is the continuously compounded rate of return in period t , then $\log(FV) = \log(PV) + r_1 + r_2$, where $\log(\cdot)$ is the natural log. The ending log value is simply the beginning log value plus the sum of all the continuously compounded periodic returns.

can be computed using the previous formulas by simply computing the present value of each future sum and then adding all these present values together. For example, suppose \$10,000 will be received in two years and another \$12,000 will be received in three years. Suppose the interest rate applicable to the payment two years from now is 10%, and the rate applicable to the payment in three years is 11% (all compounded annually). Then, the present value of the future cash flow stream consisting of these two payments is \$17,038.76, computed using formula (2) as follows:

$$PV = \frac{\$10,000}{(1 + 0.10)^2} + \frac{\$12,000}{(1 + 0.11)^3} = \$8,264.46 + \$8,774.30 = \$17,038.76$$

Of course, such a procedure gets tedious when there are many future cash flows. This is one reason it is often useful in real estate investment analysis to be able to express the present value of a multiperiod future cash flow stream using a simplified formula. Another reason is that such simplified formulas may enable interesting relationships to become apparent. For example, they can reveal the basic nature of the relationship among the current yield or income return, the expected average future growth rate of the cash flows, and the discount rate or required expected total return on a long-lived commercial property with a regular pattern of cash flow generation.

Relatively simple formulas for present value relationships can usually be found if the future cash flow stream has a regular pattern. While the cash flow streams of real buildings will not usually be so regular, they can often be well approximated (or "modeled") by regular patterns that enable simple present value formulas to be computed. These simple multiperiod formulas are the subject of this section of this chapter.

*8.2.1 Geometric Series Formula

In the real world, cash flow streams often all occur at equal intervals of time, and all change over time at a constant rate (even if that rate is zero). Such a regular stream of cash flow is known as an **annuity** if the stream has a finite life (finite number of cash flows), or a **perpetuity** if the stream is infinite. Many important real estate cash flows are of this type (such as mortgages and leases). Other real estate cash flows can be reasonably approximated by annuities or perpetuities (e.g., the net cash flow from an apartment building may be modeled as a constant-growth perpetuity). So, these are very important.

In order to explain how to value annuities and perpetuities, we think it is best to begin with first principles: a famous mathematical formula known as the **geometric series formula**. This is not just for your cultural benefit. If you understand the geometric series formula, you will have a greater depth of knowledge that will give you more flexibility to solve a wider range of real estate problems in the real world. For example, you will be able to develop a simple analytical formula for the value of a building that rents out its space in long-term leases, a typical and fundamental situation in commercial real estate. This will enable you to analyze such issues as the optimal lease term, or the trade-off between lease term and rent levels or rent growth rate. So bear with us a moment as we delve into some classical mathematics. At first this may seem complicated and abstract, but we will soon get to a very simple formula that you will be able to apply in a variety of practical real estate situations.

Suppose we have a sum consisting of N terms, each of which is exactly d times the value of the preceding term, and whose first term has the value a . Another term for d is the **common ratio**. Label the value of this sum S .

$$S = a + ad + ad^2 + \dots + ad^{N-1}$$

Now multiply this sum by the common ratio, d :

$$Sd = ad + ad^2 + ad^3 + \dots + ad^N$$

Now subtract Sd from S :

$$\begin{aligned} S &= a + ad + ad^2 + \dots + ad^{N-1} \\ -Sd &= -ad - ad^2 - \dots - ad^{N-1} - ad^N \end{aligned}$$

The intermediate terms all cancel out in the subtraction, leaving:

$$\begin{aligned} S - Sd &= a - ad^N \\ &= (1 - d)S = a(1 - d^N) \end{aligned}$$

Thus,

$$S = a(1 - d^N)/(1 - d) \quad (6a)$$

Formula (6a) is the geometric series formula. It says that the sum of the series equals the first term times the quantity one minus the common ratio raised to the power of the number of terms, all divided by the quantity one minus the common ratio.

Now note that if $d < 1$, then $d^\infty = 0$, so if the series is infinite (as in a perpetuity), then the value of the sum simplifies to

$$S = a/(1 - d) \quad (6b)$$

These simple formulas can be used to address a wide range of real estate problems. Some of the most important examples are discussed in the following sections.

8.2.2 Present Value of a Level Annuity in Arrears

Perhaps the most widely used application of the geometric series formula is to compute the present value of a **level annuity** in arrears, which is a series of equal payments made at the *end* of every period starting one period from the present. This type of cash flow stream defines the classical mortgage, with monthly payments. For example, a 30-year mortgage is a level annuity in arrears with $N = 360$.

To see how this problem is an example of the geometric formula, let's take a specific example. Suppose there are 240 payments left on a mortgage that has monthly payments of \$1,000 each. How much is the mortgage worth today if the opportunity cost of capital (discount rate) is 12% per year (actually, 1% per month) and the next payment on the mortgage is due one month from now?

We can set up this present value as the sum of a sequence of present values of single sums, using formula (2):

$$PV = \frac{\$1000}{1.01} + \frac{\$1000}{1.01^2} + \frac{\$1000}{1.01^3} + \dots + \frac{\$1000}{1.01^{240}}$$

Here, the first term is $a = \$1,000/1.01$. The number of terms is $N = 240$. And the common ratio is $d = 1/1.01$, because each term is that factor times the previous. Now if we substitute these values of a , d , and N into the geometric formula (6a) we arrive at

$$\begin{aligned} PV &= \frac{a(1 - d^N)}{1 - d} = \frac{(\$1000/1.01)[1 - (1/1.01)^{240}]}{1 - (1/1.01)} \\ &= \frac{\$1000[1 - (1/1.01)^{240}]}{0.01} = \$90,819 \end{aligned}$$

This famous case of the geometric formula is often written as

$$PV = PMT \frac{1 - 1/(1+r)^N}{r} \quad (7a)$$

or alternatively as

$$PV = PMT \left(\frac{1}{r} - \frac{1}{r(1+r)^N} \right)$$

where PMT is the level cash flow payment amount, and r is the effective simple interest rate *per payment period*.

The ratio of the present value of the annuity to the amount of each payment, PV/PMT , is often referred to as the "annuity factor," or the "present value interest factor for an annuity" (PVIFA).⁵ As we can see from formula (7a), the annuity factor is a function only of the interest rate r and the number of payment periods N : $[1 - 1/(1+r)^N]/r$. You multiply the periodic payment amount by this factor to arrive at the present value of the cash flow stream.

Keep in mind that the underlying formula that is equivalently expressed in (7a) is

$$PV = \frac{PMT}{1+r} + \frac{PMT}{(1+r)^2} + \frac{PMT}{(1+r)^3} + \dots + \frac{PMT}{(1+r)^N} \quad (7)$$

From this expansion, we can see that the annuity formula is an application of the geometric series formula, with $a = PMT/(1+r)$ and $d = 1/(1+r)$. We also see how formula (7) is built up as a series of applications of formula (2). It should be clear that an underlying assumption is that the simple effective interest rate per payment period (r) is being compounded at the end of every payment period. Thus, for example, if the payment period is monthly and $r = 1\%$, then the nominal interest rate would typically be quoted as 12%, and the effective annual rate would actually be 12.6825%, based on formula (3a).

Expressed in terms of the nominal annual rate, the annuity formula becomes

$$PV = PMT \frac{1 - \left(\frac{1}{1 + \frac{i}{m}} \right)^{(Tm)}}{\frac{i}{m}} \quad (7b)$$

where the variables are all defined as before.

8.2.3 Present Value of a Level Annuity in Advance

Sometimes the cash flows arrive at the *beginning* of each period, rather than at the end. This is particularly common for rent payments in long-term leases. For example, suppose a 20-year ground lease has a fixed net rent of \$1,000 per month paid at the beginning of the month. What is the value of this lease when it is signed if the opportunity cost of capital is 12% per year (1% per month)? We have

$$PV = \$1000 + \frac{\$1000}{1.01} + \frac{\$1000}{1.01^2} + \dots + \frac{\$1000}{1.01^{239}}$$

This is the same as the previous problem only multiplied by 1.01. Here, the first term is $a = 1,000$, but otherwise this problem is identical to the previous one (N is still 240, the number of terms on the

⁵For constant-payment mortgages, this is also known as the mortgage constant, the ratio of the mortgage payment to the initial principal loan amount.

right-hand side, or the total number of payments, and the common ratio, d , is still $1/1.01$). So, the lease is worth

$$\begin{aligned} PV &= \frac{a(1-d^N)}{1-d} = \frac{(\$1000)[1 - (1/1.01)^{240}]}{1 - (1/1.01)} \\ &= \left(\frac{1.01}{0.01} \right) \$1000 [1 - (1/1.01)^{240}] = \$91,728 \end{aligned}$$

To convert formula (7a) to give present value with payments in advance, you simply multiply the right-hand side of (7a) by the factor $(1+r)$, or equivalently by $[1 + (i/m)]$, to remove one period's discounting from the present value calculation. Thus, the annuity present value formula for payments in advance is given by formula (8a) (keeping in mind that $r = i/m$ and $N = mT$ in the usual convention).

$$PV = PMT(1+r) \frac{1 - 1/(1+r)^N}{r} \quad (8a)$$

The annuity formula is much more commonly used assuming payments in arrears than in advance. Business calculators typically have a button that allows one to set the calculator for payments either in arrears or in advance (the default setting is normally payments in arrears). Spreadsheet software formulas also typically provide an optional indicator to convert present value formulas to payments in advance, with the default being for payments in arrears.

8.2.4 Present Value of a Constant-Growth Annuity in Arrears

The annuities we have just considered were level payment annuities; that is, all their periodic cash flows were equal. Now let's relax that assumption by allowing all the periodic cash flows to grow (or shrink) at the same rate. That is, each cash flow is the same multiple of the previous cash flow. The present value of such a stream of cash flow will clearly also be an application of the geometric series formula. For example, consider a 10-year lease providing for annual payments at the ends of the years. The initial rent is \$20/SF but that rent will grow at a rate of 2% per year each year thereafter. What is the present value of the lease, per SF, assuming an opportunity cost of capital of 10% per year?

To solve this problem, first set up the underlying present value problem as a series of formula (2) type single-sums:

$$PV = \frac{\$20}{1.10} + \frac{(1.02)\$20}{1.10^2} + \frac{(1.02)^2\$20}{1.10^3} + \dots + \frac{(1.02)^9\$20}{1.10^{10}}$$

This allows us to see clearly that the first term is $a = 20/1.10$, the number of terms on the right-hand side (number of periodic payments) is $N = 10$, and the common ratio is $d = (1.02/1.10)$. Plug these values into formula (6a) to get

$$\begin{aligned} PV &= \frac{a(1-d^N)}{1-d} = \frac{(\$20/1.10)[1 - (1.02/1.10)^{10}]}{1 - (1.02/1.10)} \\ &= \$20 \frac{1 - (1.02/1.10)^{10}}{0.10 - 0.02} = \$132.51 \end{aligned}$$

So, the lease is worth \$132.51/SF.

This case of the geometric formula is referred to as the **constant-growth annuity**, and in general it is often written as

$$PV = CF_1 \left(\frac{1 - [(1+g)/(1+r)]^N}{r-g} \right) \quad (9)$$

where CF_1 is the value of the initial cash flow (periodic payment), g is the growth rate *per period* in the cash flows, and r is as always the discount rate per payment period.⁶ Note that in the special case where $g = 0$, the constant-growth annuity formula collapses to the level payment annuity formula, as it should.

8.2.5 Present Value of a Constant-Growth Perpetuity

Now consider an infinite sequence of cash flows, a perpetuity in which each periodic payment is a constant ratio times the previous payment. For example, suppose an apartment building has 100 identical units that rent at \$1,000 per month with building operating expenses paid by the landlord equal to \$500 per month. On average, there is a 5% vacancy. You expect both rents and operating expenses to grow at a rate of 3% per year (actually, 0.25% per month). The opportunity cost of capital is 12% per year (actually, 1% per month). How much is the property worth?

Once again, to solve this problem we first set it up as a fundamental series of single-sums based on formula (2). The initial monthly net cash flow is $500 \times 100 \times 0.95 = \$47,500$. Thus, the underlying present value problem is

$$PV = \frac{\$47,500}{1.01} + \frac{(1.0025)\$47,500}{1.01^2} + \frac{(1.0025)^2\$47,500}{1.01^3} + \dots$$

This allows us to see clearly that the first term is $a = 47,500/1.01$ and the common ratio is $d = 1.0025/1.01$. As this is a perpetuity, we apply the simpler version of the geometric formula (6b) instead of (6a), so we have

$$PV = \frac{a}{1-d} = \frac{\$47,500/1.01}{1 - (1.0025/1.01)} = \frac{\$47,500}{0.01 - 0.0025} = \$6,333,333$$

The building is worth \$6,333,333.

This case of the geometric formula is written in general notation as

$$PV = \frac{CF_1}{r-g}$$

with the variables defined as before, and it is referred to as the **constant-growth perpetuity formula**. (10)

8.2.6 Some Basic Economic Implications of the Perpetuity Formula

The constant-growth perpetuity formula is often taught in introductory finance as a famous and simple way to compute the fundamental value of a share in a company's common stock. The formula is referred to as the *Gordon Growth Model*, in which CF_1 is the current annual dividend payment per share, g is the expected long-term average annual growth rate in the company's dividends (presumably the growth rate in the company's earnings per share), and r is the stock market's required expected annual total return from investing in the stock (including consideration of how risky the stock is). For example, if a company currently pays \$1 per share in quarterly dividends (\$4/year), has a track record of providing consistent long-term average growth in dividends of 5% per year, and the

⁶Standard business calculators typically do not have a g key to compute growing annuities. However, you can "trick" your calculator to do this type of problem using the standard keys, by simply defining the interest rate as $1/d - 1$ [that is, $(1+r)/(1+g) - 1$]. You then enter the initial cash flow (undiscounted) as PMT , set the calculator to assume payments in advance (beginning-of-period payments, as if the problem were not in arrears, even though it is), compute the PV , and then divide the answer by $1+g$ to get it back in arrears. For example, in the previous problem, enter the interest rate as 7.84314% ($= 1.10/1.02 - 1$), enter 47,500 into PMT , 10 into N , 0 into FV , and compute $PV = 145.76$, which when divided by 1.10 will give 132.51.

market requires an average annual total return (including both dividends and capital gain) of 12%, then the stock should be worth $\$4/(0.12 - 0.05) = \$4/0.07 = \$57.14$ per share. Presumably (according to the model), if the stock is currently selling for less than that, then it is a good buy.

In general, the constant-growth perpetuity model represents a basic way to understand the value of any infinitely lived income-producing asset, as a relationship among the current level of net cash flow produced by the asset, the likely long-term average rate of growth in that cash flow, and the expected annual total return required by investors in the asset. As such, this model is broadly applicable to commercial property and provides a basic understanding of the cap rate, defined as the current annual income divided by the value of the property. For example, the constant-growth perpetuity model reveals that the apartment building described earlier would have a cap rate of 9% ($12\% - 3\%$) (this is $r - g$ on a per-annum basis). In other words, the apartment would sell for a price approximately 11 times its annual net income (that is, $1/0.09$). In fact, the building's annual net income is \$570,000 ($\$47,500 \times 12$). Dividing this by the \$6,333,333 estimated value implies the cap rate of 9%. Thus, the perpetuity model reveals that the cap rate for a building like this equals investors' expected total return minus their expected annual long-term growth rate in the building's net income.

*8.2.7 Introducing Long-Term Leases

While some types of commercial real estate, such as apartment buildings, do not make much use of long-term leases, most types of commercial property use leases that last several years or more. For example, leases in office or warehouse buildings are often 3 to 10 years long. Within each lease, once the lease is signed, the cash flows specified in the lease have relatively low risk because the tenant is contractually obligated to make the lease payments. (From the perspective of the risk of the rental cash flows, once the lease is signed, it is much like a loan to the tenant, or a bond issued by the tenant.) On the other hand, prior to the signing of any given lease, the cash flows that the building can earn under such future leases are less certain, which makes them more risky, for it is not known exactly what rent a landlord will be able to charge. Furthermore, within each lease the growth rate in the rent will often be known for certain, while between leases the growth rate will depend on conditions in the rental market.

Thus, to model the cash flows of a building subject to long-term leases, we really need to consider two different growth rates and two different discount rates. One set of rates applies *within* leases (the **intralease rates**), and the other set of rates applies *across* leases (the **interlease rates**). At first glance, this would seem to complicate the present value calculation greatly. However, the geometric series formula enables us to obtain a relatively simple formula for the present value of such a building. The present value calculation is broken down into two steps. Each lease is a constant-growth annuity whose present value as of the time of lease signing can be determined using formula (6a), and the building as a whole is a constant-growth perpetuity whose value can be determined using formula (6b). In effect, the building is a "perpetuity of successive annuities." Consider the following example.

A space will lease out in 10-year leases successively in perpetuity. The leases have annual rent payments at the beginnings of the years. Each lease will be for a constant rent once that lease is signed, but new leases will be signed at rent levels reflecting annual growth in market rent. This growth is expected to be 1% per year, but this is an uncertain, risky growth expectation. So the expected rent in each successive lease will be 1.01^{10} times the previous lease's rent, although this is not guaranteed. Once a lease is signed, the opportunity cost of capital for computing the PV of that lease as of the time of its signing is 6% per year, reflecting the relatively low risk of the contractually fixed rental payments. However, between leases or prior to lease signings, the opportunity cost of capital is 11% per year, reflecting the greater risk associated with not knowing for certain what the

future market rent will be. The initial lease is expected to be signed (and first rent received) one year from now, with an annual rent of \$10/SF. What is the value of this property per SF?

In the first step of solving this problem, we compute the present value of the first lease when it is signed, as a level annuity in advance. We have $a = 10$, $N = 10$, $d = (1/1.06)$:

$$\begin{aligned} PV(\text{Lease 1}) &= \frac{a(1 - d^N)}{1 - d} = \frac{\$10(1 - 1/1.06^{10})}{1 - 1/1.06} \\ &= \$10 \left(\frac{1.06}{0.06} \right) (1 - 1/1.06^{10}) = \$78.02/\text{SF} \end{aligned}$$

The present value of this first lease as of today is this expected value at signing (one year from now) computed above, divided by 1.11 (because this is a risky value—we do not know for sure the lease will be signed with a rent of \$10/SF). Thus, the present value of the first lease as of today is $\$78.02/1.11 = \$70.29/\text{SF}$. This is the value of the first term in the perpetuity of successive annuities that represents the present value of the property as a whole. Thus, this is the value of a in the perpetuity formula (6b) for calculating the building's value.

In the second step of solving this problem, we need to identify the common ratio, d , to apply in the perpetuity problem. In this case, $d = (1.01/1.11)^{10}$, because there are 10 years (the lease term) between each expected lease signing, and we expect market rents to grow at a rate of 1% per year in this rental market. In other words, if we let PVL_j represent the value of the j th lease at the time of its signing, then the perpetuity that represents the building value is given by the following infinite series of single-sums:

$$PV(\text{building}) = \frac{PVL_1}{1.11} + \frac{(1.01)^{10} PVL_2}{1.11^{11}} + \frac{(1.01)^{20} PVL_3}{1.11^{21}} + \dots$$

Note that the larger interlease discount rate of 11% is used here because we are discounting across leases, that is, across time prior to the signings of the future leases we are evaluating. (This is in contrast to the computation of the values, at signings, of each lease, in which the smaller intralease discount rate is applied in the annuity formula.) Thus, the value of the property is

$$PV(\text{building}) = \frac{a}{1 - d} = \frac{PVL_1/1.11}{1 - (1.01/1.11)^{10}} = \frac{\$70.29}{1 - (1.01/1.11)^{10}} = \$115.05/\text{SF}$$

So, the building is worth \$115.05/SF.

Note that this is a substantially different value than if we employed the simple constant-growth perpetuity formula to the initial \$10/SF rent level as if there were no long-term leases: $PV = CF_1/(r - g) = \$10/(0.11 - 0.01) = \$10/0.10 = \$100/\text{SF}$.

The use of the long-term leases, while delaying the growth in cash flow to only once every 10 years, reduces the risk of the cash flows sufficiently in this case so that the present value of the space is substantially greater than it would be if only short-term leases were employed. The ability to apply the underlying geometric series formula flexibly as in this problem enables one to analyze questions involving long-term leases, such as the trade-off between lease term, rental level, and building value.

8.2.8 Summarizing the Present Value of Geometric Cash Flow Streams

All of the previous problems have really only used a single underlying formula, the geometric series formula (6). The general procedure has been to set up the underlying present value problem as a sequence of single-sums, so that it is clear what is the first term on the right-hand side of the equation (the value a), what is the common ratio between those terms (d), and what is the number of terms (N). In complex problems, such as when more than one long-term lease is involved, this procedure should

be applied in more than one step, building up the overall present value problem. The following list summarizes the application of formula (6) to the basic simple cases:

- For the level perpetuity or annuity, $d = 1/(1 + r)$ where r = interest rate per period.
- For constant-growth perpetuity or annuity, $d = (1 + g)/(1 + r)$.
- For cash flows in arrears (at ends of periods), $a = CF_1/(1 + r)$.
- For cash flow in advance (at beginnings of periods), $a = CF_1$.
- For annuities, N = number of periods.
- For perpetuities, N = infinity and d must be < 1 (so, $d^N = 0$).

8.2.9 How to Convert Annuities to Future Values

All of the previous examples were present value problems in which we were converting a future cash flow stream to present value. In some cases, it is useful to do the reverse: convert a stream of cash flows into a future value as of a terminal point in time, by growing the cash flows to the terminal point. In fact, this can be done still using the same geometric formula. For example, consider the future value of an annuity of \$100/month in arrears (paid at the ends of the months) for 12 months at 12% per annum (compounded monthly, actually 1% per month). As always, the first step is to set the problem up as a sequence of single-sums based on formula (2), in this case formula (2b). Here we have

$$FV = \$100(1.01)^{11} + \$100(1.01)^{10} + \dots + \$100$$

So the first term is $a = 100(1.01)^{11}$, the number of terms is $n = 12$, and the common ratio is $d = 1/1.01$. Thus,

$$FV = \frac{\$100(1.01)^{11}[1 - (1/1.01)^{12}]}{1 - (1/1.01)} = \$1,268.25$$

In general, the future value problem simply represents $(1 + r)^N$ times the present value problem. For example, if the following gives the present value:

$$PV = \frac{CF_1}{1 + r} + \frac{CF_2}{(1 + r)^2} + \dots + \frac{CF_N}{(1 + r)^N}$$

then multiplying both sides of the equation by $(1 + r)^N$ will give the future value as in formula (11):

$$\begin{aligned} (1 + r)^N PV &= (1 + r)^N \frac{CF_1}{1 + r} + (1 + r)^N \frac{CF_2}{(1 + r)^2} + \dots + (1 + r)^N \frac{CF_N}{(1 + r)^N} \\ &= CF_1(1 + r)^{N-1} + CF_2(1 + r)^{N-2} + \dots + CF_N = FV \end{aligned} \quad (11)$$

8.2.10 Solving for the Cash Flows

If the present value can be expressed as a function of a single periodic payment amount, such as the level payment in an annuity or the initial cash flow in a growth annuity or perpetuity, then it is straightforward to invert the present value formula to solve it for the cash flow. For example, the following formula inverting (7a) tells the level payment amount (PMT) that is required to produce a given present value, given the number of payments and the interest rate:

$$PMT = PV \frac{r}{1 - 1/(1 + r)^N} \quad (7c)$$

For example, a borrower wants a 30-year, monthly-payment, fixed-interest mortgage for \$80,000. As a lender, you want to earn a return of 1% per month, compounded monthly. You agree to

provide the \$80,000 up front (i.e., at the present time), in return for the commitment by the borrower to make 360 equal monthly payments in the amount of \$822.89 each, starting with the first payment due in one month. The required monthly payment of \$822.89 is found as

$$822.89 = 80,000 \frac{0.01}{1 - 1/(1.01)^{360}}$$

8.2.11 Solving for the Number of Payments

Similarly, it is possible to invert the previous formulas to find the number of payments or length of time required for a given cash flow stream to produce a given present value. For example, to find the number of specified level payments required to pay off a given amount at a given interest rate, formula (7) can be solved for Tm :

$$Tm = -\frac{\ln(1 - \frac{i}{m} \frac{PV}{PMT})}{\ln(1 + \frac{i}{m})} \quad (7d)$$

where $\ln(\cdot)$ is the natural logarithm. In the case of the growth annuity, formula (9) is solved for N :

$$N = \frac{\ln(1 - \frac{PV}{CF_1} (r - g))}{\ln(\frac{1+g}{1+r})} \quad (9a)$$

8.2.12 Combining the Single Lump Sum and the Level Annuity Stream: Classical Mortgage Mathematics

The typical calculations associated with classical mortgage mathematics can be solved using one or the other or a combination of the single-sum formulas described in Section 8.1 and the annuity formulas just described. This is because the classical mortgage is constructed as an annuity with the contractual principal as the PV amount.⁷ To calculate the monthly payments on a loan of a given amount, you simply apply the annuity formula solved for the payment as in (7c). To calculate how much you can borrow given a specified level of monthly payments for a specified number of years, you apply the present value annuity formula (7a). To calculate the remaining outstanding loan balance (OLB) at any point in time (i.e., after a given number of payments), the annuity formula also applies. The OLB equals the present value of the remaining loan payments discounted at the loan's contractual interest rate, so you apply the annuity formula (7a) to determine the PV with N equal to the number of payments not yet made. The PV amount so determined equals the OLB on the loan. In other words, after q payments have been made on a loan that has a monthly payment amount of PMT and originally was for T years at a rate of i , the OLB is given by

$$OLB = PMT \frac{1 - (\frac{1}{1+i/m})^{(Tm-q)}}{\frac{i}{m}} \quad (7e)$$

⁷Note that the contractual principal of a loan is not necessarily exactly equal to the amount of cash the borrower will receive from the lender at the time the loan is disbursed. This is due to the possible presence of discount points of prepaid interest or loan application fees and other expenses that may be taken out of the loan disbursement. Nevertheless, the borrower's debt obligation at the time the loan is taken out equals the entire contractual principal amount.

For example, the \$80,000, 12%, 30-year mortgage we noted earlier with the monthly payments of \$822.89 would have an outstanding loan balance after 10 years of payments equal to \$74,734.39, based on the following calculation:

$$74,734.39 = 822.89 \frac{1 - 1/(1.01)^{240}}{0.01}$$

The classical mortgage is "fully amortizing," meaning that the regular monthly payments completely and exactly pay off the entire contractual principal over the course of the maturity of the loan. But many loans are not fully amortizing. They may amortize, for example, at a 30-year rate but have a maturity of only 10 years, which means that the outstanding loan balance is due after 10 years. An extreme version of this would be a loan that does not amortize at all, also known as an interest-only loan. Interest-only loans are common particularly among commercial mortgages.

In the case of partially amortizing or nonamortizing loans, the cash flow stream from the mortgage consists of the combination (or sum) of an annuity plus a single lump sum payment at the end. The annuity portion consists of the interest plus any partial amortization, while the single sum at the end consists of the remaining principal at the time of maturity of the loan. In the case of an interest-only loan, the annuity is purely interest (each payment equaling the per-period simple interest rate times the contractual principal), while the single-sum liquidating payment at the end equals the entire contractual principal amount.

Using our previous labels and formulas (i.e., FV represents the single-sum amount and PMT represents the regular periodic payment amount), a wide variety of mortgages can be represented by the following formula:

$$PV = \left(\frac{PMT}{(1+r)} + \frac{PMT}{(1+r)^2} + \cdots + \frac{PMT}{(1+r)^N} \right) + \frac{FV}{(1+r)^N} \quad (12)$$

where, as always, $r = i/m$, the simple per-period interest rate, and $N = Tm$, the total number of periods in the maturity of the loan. In formula (12), the first part on the right-hand side in the parentheses is the annuity part, and the last term on the right is the single sum. For example, a 12% interest-only loan for \$80,000 would have $PMT = \$800$ and $FV = \$80,000$, which would give $PV = \$80,000$ at the 12% nominal annual rate ($r = 12\%/12 = 1\%$), for any loan term (any value of N).

8.2.13 How the Present Value Keys on a Business Calculator Work

Business calculators typically have two sets of keys or functions that are useful in solving present value problems: the mortgage math keys and the discounted cash flow (DCF) keys. The former are designed specifically to deal with the type of level annuity and single-sum problems discussed in Section 8.2.12 and represented by formula (12). The DCF keys are designed to enable present value and **internal rate of return (IRR)** calculation of a more general stream of future cash flows. The IRR and DCF valuation will be defined and discussed in Chapters 9 and 10, while mortgages will be discussed further in Part VI. The focus of the present chapter is only on the mathematical mechanics and the functioning of the typical business calculator.

The mortgage math keys are the quickest way to solve problems that can be set up with some combination of a stream of level regular cash flows plus at most one additional lump sum cash flow at the end of the level stream. There are typically five mortgage math keys corresponding to the five variables in formula (12): PV , FV , PMT , N , and r .⁸ The mortgage math keys work by solving formula (12) for the value of the variable whose key has been pressed, given the values that currently reside in

⁸As noted, in many cases you can enter the payment frequency per year in the calculator, and then you can enter the interest in nominal annual terms, and the calculator will automatically compute $r = i/m$.

the registers of the other four variables in the formula. Note that, as per formula (12), the calculator assumes that the last one of the regular annuity cash flows whose amounts are specified in the *PMT* register occurs at the same time (*N* periods in the future) as the single lump sum amount represented in the *FV* register.

Note also that formula (12) can be rewritten to equate to zero as follows:

$$0 = -PV + \left(\frac{PMT}{(1+r)} + \frac{PMT}{(1+r)^2} + \cdots + \frac{PMT}{(1+r)^N} \right) + \frac{FV}{(1+r)^N}$$

In this form, it is seen clearly that the sign of the *PV* amount will normally be opposite to that of the *PMT* and *FV* amounts. From the lender's perspective, the *PV* amount represents an up-front cash outflow, while the other amounts are the future debt service payments that provide cash inflows. Thus, it makes sense that calculators represent amounts entered into the *PMT* and *FV* registers as being of opposite sign to amounts calculated in the *PV* register, and vice versa.⁹

8.2.14 IRR and Multiperiod DCF Formula

The DCF keys on the calculator are designed to solve a more general present value problem. If the future cash flows are not well described by the equal payments of a level annuity plus at most one lump sum at the end, then the DCF keys must be used to compute present values or internal rates of return (IRR). The DCF keys allow all the future cash flow amounts to be different.

Consider an arbitrary stream of cash flows occurring at the ends of each of a consecutive sequence of uniform-length time periods (net cash flow amounts labeled CF_0 , CF_1 , CF_2 , etc., occurring in time periods labeled 0, 1, 2, and so on, where period 0 is the present). These cash flow amounts are entered into the calculator cash flow registers, which are often labeled something like CF_j or $C0$, $C1$, and so on. The DCF keys solve the equation in formula (13) for either the *NPV* value or the *IRR* value:

$$NPV = CF_0 + \frac{CF_1}{(1+IRR)} + \frac{CF_2}{(1+IRR)^2} + \cdots + \frac{CF_T}{(1+IRR)^T} \quad (13)$$

where *T* is the number of future periods encompassed in the analysis (beyond the present).¹⁰

To solve for the *IRR*, the *NPV* is by definition assumed to be zero, and the amount in the CF_0 register must be of opposite sign to the bulk of the values in the other CF_j registers ($j = 1, \dots, T$).

⁹An exception is when using the mortgage math keys to compute the terminal value (*FV*) of a level stream, as in formula (11). For such problems, set *PV* = 0 in the calculator and solve for *FV*. The *FV* amount represents the terminal (compound growth) value of the payments, with *FV* being opposite in sign to *PMT*, as in formula (11) set equal to zero. Note that in general there is no algebraic analytical formula to solve formula (13) for the *IRR*. The calculator must find the answer by a repetitive trial-and-error process through which it iterates to an answer that is acceptably close to solving the equation.

¹⁰Note that equation (13) is the most general formula for the present value of an arbitrary stream of discrete future cash flows assuming a constant discount rate. (If the cash flows occur at irregularly spaced points in time, one can simply define the smallest common time divisor as the period, and have $CF_i = 0$ between the cash flow receipts.) An even more general formula would allow for time-varying discount rates, as follows:

$$NPV = CF_0 + \frac{CF_1}{(1+r_1)} + \frac{CF_2}{(1+r_2)^2} + \cdots + \frac{CF_T}{(1+r_T)^T}$$

where r_t is the time-weighted average multiperiod return from time zero to time *t*. It is useful to note that even in this most general form, the present value equation is "homogeneous of degree one" in the cash flows. That is, if all the cash flow amounts on the right-hand side of the equation are multiplied by any constant *a*, then the left-hand side *NPV* is also multiplied by that same constant. More formally: if $NPV = f(CF)$, then $aNPV = f(aCF)$, where $f(\cdot)$ is the present value equation, *CF* here is an arbitrary vector of cash flows, and *a* is a scalar constant. This is simply a consequence of the distributive law of multiplication and addition: $a(b+c) = ab+ac$, but it allows great computational convenience in working with present values. For example, the ability to use tables of "annuity factors," and the ability of lenders to publish loan interest rates applicable to loans of varying amounts, is a consequence of this homogeneity property.

For example, you would typically enter the up-front investment amount as a negative number representing cash outflow in register CF_0 , such as the price of the property, while the subsequent cash flows would be largely positive representing the net cash dividends or rental payments received from the investment. If the discount rate were then specified (typically in the interest key of the calculator), then this discount rate would be used in place of the *IRR* rate in formula (13) and the formula would be solved to determine the net present value (*NPV*) of the project. Alternatively, if you press the *IRR* register, the calculator will compute the value of the *IRR*.¹¹ Finally, if you specify an amount of zero for the CF_0 register, then pressing the *NPV* key will compute the present value (corresponding to the *PV*, or gross present value) of the future stream of cash flows (in arrears). Note that in this case the *NPV* register does not in itself compute a *net* present value, as we have 0 in the present cash flow.

8.3 CHAPTER SUMMARY

This chapter has presented the major formulas and procedures that are most useful for the calculation and analysis of the present value of the types of cash flow streams most commonly found in commercial real estate. Included in this are the important inverses of the present value formulas, that is, formulas for calculating implied returns, terminal values, and the time (or number of payments) until a certain present or future value is achieved. We have focused on the formal mechanics of the basic formulas, leaving the economic reasoning and analysis necessary for the correct application of these formulas largely to be covered in later chapters.

KEY TERMS

Single-sum formulas
present value
future value
discounting
growing
discount rate
interest rate
compounding
growth rate
compounding frequency

effective annual rate (EAR)
equivalent nominal annual rate (ENAR)
mortgage equivalent rate
bond equivalent (coupon equivalent) rate
continuous compounding
annuity
perpetuity
geometric series formula

common ratio
level annuity
constant-growth annuity
constant-growth perpetuity
intralease rate
interlease rate
internal rate of return (IRR)
calculator registers
(N,I,PV,PMT,FV,NPV,IRR,CF)

STUDY QUESTIONS

- What is the present value of an offer of \$15,000 one year from now if the opportunity cost of capital (discount rate) is 12% per year simple interest?
- What is the present value of an offer of \$14,000 one year from now if the opportunity cost of capital (discount rate) is 11% per year simple interest?

¹¹In some calculators, the *IRR* may be reported on a per-period basis, so that, for example, if the periods were months, you would have to multiply the computed *IRR* by 12 to obtain the nominal per-annum rate.